

Member ID: _____

Time: _____

Rank: _____



SQL DATABASE FUNDAMENTALS (345) **NATIONAL 2023**

Multiple Choice:

50 @ 2 points each

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. What output would the following query produce – SELECT "Hello from BPA" FROM Customers WHERE (TRUE OR !TRUE) AND NOT(!FALSE) OR TRUE?
 - A. No result
 - B. Hello from BPA
 - C. Hello from BPA repeated for every record in the Customer table
 - D. Hello from BPA, but only from the first record
2. What does the acronym OLAP stand for?
 - A. Ordered linear programming
 - B. Online analytical processing
 - C. Organizational lists advanced programming
 - D. Object language advanced package
3. To represent the relative compliment of two tables in SQL you would use a(n) _____ join.
 - A. OUTER
 - B. INNER
 - C. FULL
 - D. COMPLIMENT
4. Given the table mytable, where field1 is either 1 or 0 and field2 is either 1 or 0, what will be the result of the query “SELECT field1^field2 FROM mytable”?
 - A. A 0 for each row where field1 and field2 are 0; all other rows will be 1
 - B. Only rows from mytable where either field1 or field2 is 1
 - C. A 0 where both field1 and field2 have the same value; all other rows will be 1
 - D. Only rows where either field1 is 1 or field2 is 1, but not both
5. The result of a(n) _____ join is known as a Cartesian Product.
 - A. FULL
 - B. CROSS
 - C. OUTER
 - D. INNER
6. The WHERE and HAVING clauses can be used interchangeably – the WHERE clause processes data during the QUERY and the HAVING clause processes data once the QUERY has finished.
 - A. TRUE
 - B. FALSE

7. To achieve the normalization standard of _____, you must eliminate any transitive dependencies.
 - A. 3NF
 - B. BCNF
 - C. 4NF
 - D. 5NF
8. A _____ key is a combination of one or more columns that can uniquely identify records in a table.
 - A. Foreign
 - B. Combined
 - C. Complex
 - D. Composite
9. Which of the following is an appropriate step to take to lessen the potential of an SQL injection attack on your database?
 - A. Implement brute force controls such as account lockout after a set number of invalid attempts
 - B. Require users change their passwords regularly
 - C. Use parameterized queries instead of dynamic queries
 - D. Do not house the database and application servers on the same physical machine
10. _____ is a condition where the foreign key column values in all of the rows of one table have matching rows in the primary key of a second table.
 - A. Restriction factor
 - B. Record Instance
 - C. Referential integrity
 - D. Concurrency
11. The _____ property ensures that all operations within a transaction are successful.
 - A. Atomicity
 - B. Consistency
 - C. Completeness
 - D. Durability
12. GRANT and REVOKE are which types of commands?
 - A. DCL
 - B. DDL
 - C. TCL
 - D. DML
13. A clustered index _____.
 - A. modifies the original table to sort in place
 - B. creates a separate file that hold a key and its position
 - C. must be created on the primary key
 - D. is only available for tables with 3 columns or less

14. _____ is a procedure in DBMS which manages two simultaneous processes to execute without conflicts between each other.
- A. Dependency management
 - B. Concurrency control
 - C. UTLI
 - D. Strobe control
15. Which of the following is NOT a Data Model in a DBMS?
- A. Hierarchical Model
 - B. Network Model
 - C. Relational Model
 - D. Functional Model
16. CREATE, ALTER, & RENAME are all examples of which data language?
- A. DDL
 - B. DCL
 - C. DML
 - D. DXL
17. Assume a query fails in the middle of a transaction and an auto rollback takes place. Which of the following describes why the auto rollback takes place?
- A. Durability
 - B. Isolation
 - C. Atomicity
 - D. Transactional persistence
18. Assume a transaction completes and then there is a system failure. Which of the following describes why the data persists after a system failure?
- A. Durability
 - B. Isolation
 - C. Atomicity
 - D. Transactional persistence
19. What would be the result of the following query?
- SELECT 5+5 where 4 not in (1, 2, NULL)
- A. 10
 - B. Syntax Error
 - C. No Rows
 - D. 55

Consider the following three tables for questions 20-27

Member's m_id is auto-increment, unique; Item's id is auto-increment, unique; All fields are non-null

customer			member_info		sales			
cust_no	mem_no	title	mem_no	name	mem_no	month	gross	net
1164	1200	Boss	1200	Tom Thumb	1500	Jan	1000.0	900.0
1781	1000	Level 1	1000	Jingle Bells	1500	Feb	1000.0	905.0
1531	1500	Level 1	850	Boss Lady	5431	Jan	500.0	450.0
1782	850	Level 2	5431	Danger Dan	1500	June	1000.0	925.0
1414	5431	Level 1	1500	Sally Speed	850	Jan	500.0	450.0
					1200	Feb	750.0	750.0
					850	Apr	500.0	425.0
					1500	Oct	1000.0	925.0
					850	Mar	500.0	250.0

20. The highest normalization form the member_info table meets is _____.
 A. 2NF
 B. BCNF
 C. 4NF
 D. 5NF
21. When the customer table was created the cust_no was set up as a primary key and the mem_no was not created as a key. Both pieces of data uniquely identify each given record. If the mem_no was added a key, which type of key would it be?
 A. Foreign
 B. Candidate
 C. Associate
 D. Primary
22. How would the key above be created on the customer table?
 A. ALTER TABLE customer ADD UNIQUE KEY cust_no(cust_no)
 B. ALTER TABLE customer ADD KEY cust_no
 C. ALTER TABLE customer ADD PRIMARY KEY cust_no(cust_no)
 D. ALTER TABLE customer MODIFY KEY mem_no(mem_no, cust_no)

23. The sales table above was originally created in error with the following create statement. What statement had to be run before the data was inserted?

```
CREATE TABLE `sales` (  
  `mem_no` INT NOT NULL,  
  `month` VARCHAR(45) NULL,  
  `gross` DOUBLE NULL,  
  `net` DOUBLE NULL,  
  PRIMARY KEY (`mem_no`));
```

- A. ALTER TABLE sales MODIFY mem_no ALLOW DUPS
 - B. ALTER TABLE sales MODIFY COLUMN mem_no INT NULL
 - C. ALTER TABLE sales DROP PRIMARY INDEX
 - D. ALTER TABLE sales MODIFY INDEX ("cust_no") REMOVE PRIMARY CONSTRAINT
24. The highest normalization form the customer table meets is _____.
A. 2NF
B. BCNF
C. 4NF
D. 5NF

25. How many rows would the following query produce?

```
SELECT name  
  FROM member_info a  
        INNER JOIN sales b ON a.mem_no = b.mem_no  
 WHERE a.mem_no IN  
        (SELECT mem_no  
          FROM sales  
          GROUP BY mem_no  
          HAVING SUM(gross) > 1500)
```

- A. 0
- B. 1
- C. 3
- D. 5

26. Why will this query result in a syntax error?

```
SELECT name
  FROM member_info
 WHERE mem_no NOT IN
      (SELECT mem_no
        FROM sales
       GROUP BY month
      WHERE AVG(net) > 500)
```

- A. Cannot use an aggregate function in a WHERE clause
- B. Cannot have two WHERE clauses in the same query
- C. NOT is not allowed in this context
- D. GROUP BY column is not part of the field list

27. The customer and member_info table appear to have a one to one relationship. What would be the need to have the tables separated?

- A. Having them together violates BCNF
- B. Having them together violates 4NF
- C. Having them together violates 5NF
- D. There is no need to have them as separate tables

28. A WITH clause is used in SQL to support CTSs. What does CTE stand for?

- A. Common Table Expression
- B. Concurrent Tabulation Entity
- C. Consistent Tab Execution
- D. Classic Transposition Encoding

29. Using a WITH clause, the temporary entity remains _____.

- A. for the duration of the current transaction
- B. until it is manually deleted
- C. for the duration of the current database connection
- D. for the duration of the query it is a part of

30. With using a stored procedure, SQL queries can still be recursive.

- A. True
- B. False

31. In SQL the EXCEPT clause is closely related to what other clause?

- A. WITH
- B. ALTER
- C. UNION
- D. NOT

32. Which does the acronym BASE refer to in database development?

- A. Branching, Available, Structured, Enterprise
- B. aBsolute structure, Active monitoring, Syntactic, Entity relationship
- C. Basically Available, Soft state, Eventually consistent
- D. Broad Action, Strong relationships, Extensible content

33. The BASE Model of data manipulation is used in place of the ACID model for which type of Database?
- A. PostgreSQL
 - B. MongoDB
 - C. MariaDB
 - D. Microsoft SQL Server
34. _____ is what happen if a database table is locked exclusively and a process is stopped from accessing the table until the lock is release.
- A. A deadlock
 - B. Blocking
 - C. Impeding
 - D. Stopping
35. _____ is what happens when a transaction has to wait for an indefinite amount of time to acquire a lock.
- A. Increasing priority
 - B. Starvation
 - C. Wait-die and wound
 - D. Resource leak
36. One benefit of a NoSQL database over a Relational Database is _____.
- A. security and compliance
 - B. data integrity
 - C. the strength of the transaction
 - D. the relative ease of updating the schema
37. When designing a database, _____ is one reason to not fully normalize some of the data.
- A. data security
 - B. query efficiency
 - C. atomicity
 - D. requires less disk space
38. One method of ensuring the database only has data in that that belongs is to use _____.
- A. constraints
 - B. error logging
 - C. stored procedures
 - D. audit transformation
39. Given two resources and two users, each user wants both resources. If user1 has resource 1 locked and user 2 has resource 2 locked, this situation is known as a _____.
- A. hot lock
 - B. lock timeout
 - C. stored procedures
 - D. dead lock

40. A(n) _____ can occur after or instead of a DML action.
- A. update
 - B. alter
 - C. trigger
 - D. transaction
41. The highest of the four levels of the isolation property is known as _____.
- A. read committed
 - B. serializable
 - C. absolute
 - D. immediate
42. Database _____ is the ability of a database to allow multiple users to affect multiple transactions.
- A. synchronicity
 - B. concurrency
 - C. multiplicity
 - D. concomitants
43. A _____ occurs where a transaction is allowed to read a row that has been modified by another transaction.
- A. phantom read
 - B. dirty read
 - C. lost read
 - D. non-repeatable
44. A(n) _____ occurs when two transactions want to change the same column and the second update overwrites the first.
- A. lost update
 - B. optimistic transaction
 - C. phantom update
 - D. overwrite anomaly
45. An advantage of a database view is it _____.
- A. allows for easier database updates
 - B. reduces the dependency on pessimistic locks
 - C. provides consistency to applications
 - D. allows for quicker updates and reads
46. A(n) _____ defines the conceptual view of a database.
- A. Composite Attribute
 - B. Mapping Cardinality
 - C. ER Model
 - D. Relational Set

47. Which of the following is NOT a benefit of referential integrity?
- A. Ensures no duplication of data
 - B. Allows orphan data to be found and flagged
 - C. Avoids one table pointing to non-existent field in another table
 - D. Prevents the deletion of a record that is reference by a foreign key in another table
48. A _____ database is a data model in which records are organized into a tree-like or parent-child structure.
- A. hierarchical
 - B. cloud
 - C. distributed
 - D. centralized
49. Which of the following is NOT a type of OLAP server?
- A. Specialized SQL servers
 - B. Relational OLAP
 - C. Distributed OLAP
 - D. Hybrid OLAP
50. IBM devised a system of the four Vs to stand for the four dimensions of big data. What are they?
- A. Volume, Velocity, Variety, Veracity
 - B. Veracity, Variable, Virtual, Volume
 - C. Virtual, Value, Volume, Vantage
 - D. Volume, Virtual, Variable, Veracity